AVINASH — 3-Minute College Presentation Script

Territory Runner | Subsystem Focus: Algorithms & Master Documentation

TOTAL TIME: 3:00 MIN

ROLE: ALGORITHMS & DOCS LEAD

PRESENTER	ASSIGNED TIME	CORE STACK
AVINASH	Minute 6:00 to 9:00 (3 Mins)	Clipper2, Shoelace Math, Jordan Curve, Docs

⏱️ MINUTE 1 (6:00 – 7:00) : Lasso Loop Closure Algorithm

6:00 - 7:00

"Good morning professors and examiners. My name is **Avinash**. In *Territory Runner*, I was responsible for the **computational geometry algorithms, cross-platform mathematical parity, and master project documentation**.

When an athlete runs outdoors, they rarely return to their exact millimeter starting coordinate. Instead, they run an organic loop and cross their own historical breadcrumb track. I formulated our **Lasso Closure Algorithm**.

To prevent GPS drift from creating false loops, closure detection activates only after running **150 meters**. When the current coordinate comes within a **25-meter proximity buffer** of any past segment, the algorithm calculates the exact line segment intersection point, slices the closed polygon ring, and discards the dangling 'lasso tail' in real time using Clipper2."

> 👉 ***Action:*** *Switch to Slide 8. Point at the lasso loop closure diagram showing track crossing and virtual intersection point P*.*

⏱️ MINUTE 2 (7:00 – 8:00) : Algorithmic Parity & Donut Hole Classifier

7:00 - 8:00

"A fundamental theoretical problem was **algorithmic parity** between client and server. Evaluating complex ellipsoidal geodesics on mobile UI threads causes UI stutters. I implemented a scaled planar **Shoelace formula** on Flutter, correcting for latitude distortion by scaling longitude differences with the cosine of the mean latitude.

To guarantee fairness, I established **15 cross-platform golden test fixtures**. We mathematically proved that the client's Shoelace preview and the server's PostGIS `ST_Area` calculation match with **less than 0.5% variance** across all shapes.

Additionally, I authored our **Ray-Casting Donut Hole Classifier** based on the Jordan Curve Theorem. If an athlete runs around an obstacle like a lake, semi-infinite horizontal rays determine whether enclosed rings are exterior land or inaccessible interior holes."

> 👉 ***Action:*** *Switch to Slide 9. Highlight the Shoelace formula, the 0.5% parity proof, and the ray-casting visual.*

⏱️ MINUTE 3 (8:00 – 9:00) : Master Documentation, Golden Verification & Conclusion

8:00 - 9:00

"On the documentation front, I compiled our comprehensive **Software Requirements Specification (SRS), architectural sequence diagrams, and API contract specifications**, ensuring clear module boundaries across our team.

Our geometry engine is verified by **15 golden geospatial test suites** running in both Dart and Go, covering figure-8s, tiny loops, cycling routes, and pause/resumes.

In conclusion, Territory Runner proves that server-authoritative distributed systems, high-contrast athletic UI,

and rigorous computational geometry can unite into an exceptional fitness platform. My teammates Ayush, Gurudutt, and I are now ready for your questions."

> 👉 ***Action:*** *Switch to Slide 10. Display the summary dashboard, invite questions, and stand together as a team.*

AVINASH — Slide Deck Presentation Guide

Exactly what to display and point at during your 3 minutes

3 SLIDES TOTAL

SLIDE 8 (6:00 - 6:50) — **LASSO ALGORITHM**

Lasso Loop Closure Snapping

- Self-crossing detection: runner crossing their own track breadcrumbs.
- **Activation Guards:** Distance > 150m, Snap radius $R_{snap} = 25\text{m}$.
- Line intersection math: $P = A + t(B - A) = C + u(D - C)$.
- Automatic polygon ring extraction & dangling tail pruning.

SLIDE 9 (6:50 - 7:40) — **GEOMETRIC PARITY**

Shoelace vs PostGIS & Donut Holes

- Scaled planar Shoelace on mobile: $A = \frac{1}{2} |\sum (x_i y_{i+1} - x_{i+1} y_i)|$.
- **Strict Parity:** < 0.5% area discrepancy against PostGIS `ST_Area`.
- Ray-Casting Donut Classifier: Jordan Curve Theorem detecting interior voids.

SLIDE 10 (7:40 - 8:30) — **DOCUMENTATION & SUMMARY**

Master Project Documentation & Team Verification Metrics

- Complete Software Engineering Documentation: SRS, Architecture Diagrams, API Contracts, and Viva Defense Guide.
- 15 Cross-Platform Golden Geospatial Fixtures passing in both Dart and Go.
- **Combined Platform Health:** 317 Flutter tests passing + 79 Go tests passing + 0 compiler warnings.

📋 Avinash's Quick Reference Pocket Card

CODE & FILES

- **Lasso Engine:** `lib/features/gameplay/polygon_enclosure_engine.dart`
- **Shoelace Geodesy:** `backend/internal/geo/shoelace.go`
- **Donut Hole Classifier:** `lib/features/gameplay/donut_hole_classifier.dart`
- **Golden Test Vectors:** `test_fixtures/golden_valid_loop.json` (15 fixtures total)
- **Parity Boundary:** Area discrepancy strictly < 0.5% across all golden cases.

AVINASH — Examiner Viva Defense Cheat Sheet

Top 4 killer examiner questions with rapid 15-second oral defenses

EXAMINER DEFENSE

Q1: What is the time complexity of the Lasso Loop Closure detection algorithm?

15-Second Defense: "Naive intersection testing is $O(N^2)$, which freezes phones with thousands of waypoints. We optimize it to $O(M)$ by only testing the latest GPS segment against historical segments within our 25-meter bounding box. This reduces candidate checks to fewer than 20 segments, running in under 0.2 milliseconds at 60 FPS."

📁 Implemented in: `lib/features/gameplay/polygon_enclosure_engine.dart`

Q2: Why does the Planar Shoelace formula work on Earth's curved spherical surface?

15-Second Defense: "Because human athletic runs are localized ($< 5\text{ km}$ across), Earth's curvature error is negligible (<0.2%) when longitude is scaled by the cosine of the mean latitude. Across our 15 golden fixtures, the area difference between client Shoelace and server PostGIS ellipsoidal geodesics is proven to stay strictly below 0.5%."

📁 Implemented in: `backend/internal/geo/shoelace.go`

Q3: How does the Jordan Curve Theorem work in your Ray-Casting Donut Hole Classifier?

15-Second Defense: "The Jordan Curve Theorem proves any closed planar loop divides space into inside and outside. We cast a semi-infinite horizontal ray from a candidate interior point. If it crosses the polygon boundary an odd number of times, the point is inside; an even count means outside. This allows us to exclude lakes and buildings from territory claims."

📁 Implemented in: `lib/features/gameplay/donut_hole_classifier.dart`

Q4: How did you structure the software documentation and cross-platform verification fixtures?

15-Second Defense: "I structured the documentation into clear SRS specifications, architectural data flows, and 15 standardized JSON golden test fixtures. Both our Dart and Go test suites ingest these identical JSON files, ensuring that any algorithmic change in Go or Flutter is verified against the same ground truth."

📁 Implemented in: `test_fixtures/` & `docs/VERIFICATION.md`

Candidate Signature	Internal Project Guide	External Viva Examiner
AVINASH	Dept. of CSE	University Board